Define 3 types of capital.

- Regulatory required capital
- Available capital
- Target capital

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

- **Regulatory required capital** – Required for p/h protection
  - Regulators represent p/h interests (solvency focused)
  - Held in safe investments for adverse events
  - Min thresholds trigger regulator intervention
- **Available capital** – Used for growth, operational efficiency, or competitive positioning
- **Target capital** – Reflects stakeholder perspectives:
  - Stockholders: Seek gains; prefer less conservatism
  - Debtholders: Prefer conservative capital
  - Regulators/p/h: Prefer conservative capital
  - Management: Balance; avoid excessive capital

Compare factor-based vs. model-based regulatory capital methods.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

1. **Factor-based** (US)
   - RC requirements = factors $\times$ financial statement amounts
   - Factors sometimes set using industry-wide models
   - Aggregate RC = individual RCs combined via correlation matrix
2. **Model-based** (Canada, EU, Bermuda)
   - More actuary judgment
   - Models subject to regulatory review/approval

Describe NAIC RBC and primary risk categories.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

NAIC: Mostly factor-based with some model-based components

- Many assumptions prescribed
- Shift to principles-based → more model-based
- Assets/liabilities measured under statutory accounting (conservative, mostly BV-based)

4 primary risk categories:

1. C0/C1: Asset Risk
2. C2: Insurance Risk
3. C3: Interest Rate/Market Risk
4. C4: Business Risk

Describe NAIC RBC asset risk component (C0, C1).

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

- $C0$: Default risk for affiliated companies, OBS items (derivatives)
- $C1$: Divided into $C1cs$ (unaffiliated common stock) and $C1o$ (all other)
- $C1o$: Default risk for bonds, mortgages, RE, preferred stock
  - RC = factor $\times$ BV of asset
  - Factors $\uparrow$ as credit rating $\downarrow$ (0% for Treasuries, 30% for junk bonds)
- Additional adjustments:
  - Asset concentration factor $\times$ 10 largest holdings
  - Diversification factor $\times$ total bond RC (factors $\downarrow$ as issuer count $\uparrow$)

Describe NAIC RBC insurance risk component (C2).

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

- $C2_{\text{mort}}$: Risk DBs $>$ expected
  - RC = factor $\times$ NAR
  - Factors vary by:
    - Individual vs. group life
    - Insurer's ability to change NGEs
    - Portfolio size (factor $\downarrow$ as inforce $\uparrow$)
- $C2_{\text{long}}$: Risk payout annuity benefits $>$ expected
  - RC = factor $\times$ annuity reserves
  - Only life-contingent annuities in active payout
  - Factors vary with reserve size
- Final C2 reflects negative correlation between mortality/longevity:

$$C2 = \sqrt{C2_{\text{mort}}^2 + C2_{\text{long}}^2 + 2 \times C2_{\text{mort}} \times C2_{\text{long}} \times \text{CorrFactor}}$$

Describe NAIC RBC interest rate/market risk component (C3).

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

- $C3a$: Interest rate change risk (factor $\times$ reserves)
- $C3b$: Health care capitation risk
- $C3c$: Market risk
  - C-3 Phase I: Certain fixed annuities, single premium life
    - Model-based, stochastic projection with prescribed scenarios
    - RC based on subset of worst scenarios
  - C-3 Phase II: VAs
    - Model-based, stochastic projection with prescribed scenarios
    - Requires <u>reserves</u> at CTE 70
    - RC = CTE 98 Reserves $-$ CTE 70 Reserves
  - Non-single premium life RC = factor $\times$ reserves

Describe NAIC RBC business risk component (C4).

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

- Covers OR and any other risk not in C0–C3
- RC = factor $\times$ annual premiums (premiums gross of reinsurance)

Formula for NAIC authorized control level RBC and ratio?

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

Authorized control level (ACL) RBC = covariance-adjusted sum

$$\text{ACL RBC} = C0 + C4a + \sqrt{(C1o + C3a)^2 + (C1cs + C3c)^2 + C2^2 + C3b^2 + C4b^2}$$

$$\text{RBC ACL Ratio} = \frac{\text{Total Adjusted Capital}}{\text{ACL RBC}}$$

- $C0$ = Asset risk (affiliated amounts)
- $C1cs$ = Asset risk (unaffiliated common stock)
- $C1o$ = All other asset risk (excl. common stock)
- $C2$ = Insurance risk
- $C3a$ = Interest rate risk
- $C3b$ = Health credit risk
- $C3c$ = Market risk
- $C4a$ = Business risk (non-health)
- $C4b$ = Health admin expense business risk

List US regulatory action levels.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

Level	**RBC ACL Ratio**	**Action**
Trend Test Corridor	[200%, 300%)	Company performs trend test
Company Action	[150%, 200%)	Company prepares/submits RBC plan to regulator
Regulatory Action	[100%, 150%)	Company submits RBC plan; regulator issues corrective action order
Authorized Control	[70%, 100%)	Regulator authorized to take any actions necessary
Mandatory Control	$<$70%	Company deemed insolvent, placed under regulatory control

Describe Canadian solvency regulation and available capital tiers.

Canadian insurers regulated at federal level by OSFI

- Canadian assets/liabilities based on IFRS

LICAT – significant capital assessment measure used by OSFI

Available Capital = Tier 1 Capital + Tier 2 Capital

- Tier 1 Capital = mostly shareholder equity, retained earnings (no limit)
- Tier 2 Capital = assets not meeting Tier 1 criteria
  - Hybrid instruments, subordinated debt
  - Tier 2 must be positive and $\leqslant$ Tier 1

Write Canadian LICAT ratio formulas.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

$$\text{TR} = \frac{\text{Available Capital} + \text{Surplus Allowance} + \text{Eligible Deposits}}{\text{Base Solvency Buffer}}$$

$$\text{CR} = \frac{\text{Tier 1 Capital} + 70\% \text{ of Surplus Allowance} + 70\% \text{ of Eligible Deposits}}{\text{Base Solvency Buffer}}$$

- **Total Ratio**: Focuses on p/h/creditor protection
- **Core Ratio**: Focuses on financial strength
- **Surplus Allowance** = PfADs calculated under:
  - CALM for pre-2023 business
  - IFRS 17 for business issued 2023+
- **Eligible Deposits** = Collateral/LOCs from unregistered reinsurers + claims fluctuation reserves
- **Base Solvency Buffer (BSB)** = 1-year CTE 99 min capital requirement

How is the Base Solvency Buffer determined under LICAT?

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

BSB for <u>each</u> of 6 geographic regions = sum of 5 risk components:

1. Asset default risk (credit + market risk)
2. Mortality/morbidity/lapse risks
3. Interest rate risk
4. Segregated funds risk
5. Foreign exchange risk

Each item calibrated to 1-year CTE 99 level

Additional adjustments also apply:

- Par products, diversification, reinsurance, collateral, hedges

List Canadian regulatory action levels.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

	Target	Minimum
Total Ratio	100%	90%
Core Ratio	70%	55%

- If either ratio approaches target → OSFI assesses, takes necessary actions
- If either ratio $<$ minimum → OSFI can take control
- Companies must hold min \$5M available capital regardless

Describe capital requirement levels under Solvency II.

EIOPA sets standards for EU members, including Solvency II guidance

Solvency II establishes 2 capital requirement levels:

1. **Solvency Capital Requirement (SCR)**: 1-year 99.5% probability
    - Based on Basic Solvency Capital Calculation or internal model
    - Internal models subject to regulator approval
2. **Minimum Capital Requirement (MCR)**: Level requiring regulator intervention
    - Set at 1-year 85% solvency probability

Describe asset/liability valuation under Solvency II.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

- Stresses starting economic balance sheet (similar to LICAT)
- Assets/liabilities are **marked to model**
  - Model calibrated rather than directly from market data
- Liabilities based on **exit value**
  - Based on transfer/settlement value between 2 willing parties with equal information
  - Difficult to determine: No trading exchanges, sparse M&A activity
  - Often determined using discounted cash flow models
    - Discount rate = risk-free rate with adjustments

Describe key components of Solvency II balance sheet.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

- **Assets = Assets Backing Tech. Provisions + Basic & Auxiliary Own Funds**
  - Assets Backing Technical Provisions = All Tier 1 Assets + Some Tier 2 Assets
    - Assets Backing Technical Provisions = Technical Provisions (liability side)
  - Basic/Auxiliary Own Funds = Remaining Tier 2 Assets + Tier 3 Assets
- **Liabilities & Surplus = Technical Provisions + ECR + Free Own Funds**
  - Technical Provisions = BEL + Risk Margin
  - Enhanced Capital Requirement (ECR) = SCR + MCR
  - Free Own Funds = any additional surplus assets held

    = Assets − (Technical Provisions + ECR)

Basic own funds currently exist; auxiliary could come from parent

Describe MCR’s relationship to Basic SCR under Solvency II and SCR action levels.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

Basic SCR reflects correlations across/within risk categories:

$$\text{Basic SCR} = \sqrt{\sum_{i,j} \text{Corr}_{i,j} \times \text{SCR}_i \times \text{SCR}_j}$$

$$\text{SCR Ratio} = \frac{\text{Available Capital}}{\text{SCR}}$$

MCR must be between 25% and 45% of Basic SCR and $\geqslant$ 1.2M

Describe Solvency II capital tiers.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

Tier 1 = highest quality; Tier 3 = lowest

- At least 50% of SCR must be backed by Tier 1
- Max 15% of SCR can be backed by Tier 3
- At least 80% of MCR must be backed by Tier 1 capital, <u>no Tier 3</u>

Describe regulatory action levels under Solvency II.

Company approaching MCR must submit plan to regulator

- Regulator must approve plan
- If capital $<$ SCR or MCR $\rightarrow$ regulator requires more capital
- EU regulators have latitude to decide further actions

Regulatory actions determined by SCR Ratio:

Level	SCR Ratio	Action
Early Intervention Process	[25%, 45%)	Increasingly severe prescribed supervisory actions as capital → 25%
Minimum Required Level	$<$25%	Company loses authorization to operate

List Solvency II risk categories.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

1. **Underwriting risks**
   - Non-life (P&C)
   - **Life** (7 categories)
   - Health
2. **Market risks**
   - 2.1 Interest rate risk
   - 2.2 Equity Risk
   - 2.3 Property risk (RE)
   - 2.4 Spread risk (credit risk)
   - 2.5 Market risk concentration
   - 2.6 Currency risk
3. **Counterparty default**
   - Type 1: All risk mitigation exposures (e.g., reinsurers)
   - Type 2: Future receivables from p/h, mortgage loans

List Solvency II life UW risk subcategories and shocks.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

1. **Mortality risk**: Life ins mortality +15%
2. **Longevity risk**: Payout annuity mortality -20%
3. **Disability-morbidity risk**: Stresses initial claims, time on claim
4. **Lapse risk**: RC = worst of 3 stress tests:
    - +50% permanent lapse shock
    - -50% permanent lapse shock
    - -40% PIF reduction
5. **Expense risk**: Expenses +10% all years + inflation +1%
6. **Revision risk**: Payout annuity benefits +3%
7. **Catastrophe risk**: Add 0.15% to 1-year mortality rate

All calcs based on FMV assumptions with possible adjustments:

- Spread adjustment: Accounts for “irrational” market movements
- Matching/volatility adjustment: Can be made to discount rates

Describe Bermuda regulatory balance sheet.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

Capital requirements prescribed by **Insurance Act**, regulated by **BMA**

- **Assets = Assets Backing EBS Technical Provision + Surplus Assets**
- **Liabilities + Surplus = EBS Technical Provision + ECR + Free Surplus**
  - EBS Technical Provision = BEL + Risk Margin
  - Enhanced Capital Requirement (ECR) = max[BSCR, MSM]
    - Expressed as MSM + Required Capital Over MSM
    - BSCR calibrated to 99.5% solvency (similar to Solvency II)

Minimum Margin for Solvency (MSM) = max of:

1. 25% $\times$ ECR
2. \$1M BMD for Class 3A/3B insurers or \$100M BMD for Class 4 insurers

MSM available capital based on “Bermuda Statutory” financials (US GAAP, IFRS)

Describe target capital level under Bermuda regulation.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

Target Capital Level (TCL) = 120% $\times$ ECR

- TCL = amount of capital insurers expected to hold
- TCL available capital based on economic balance sheet (EBS)
- EBS assets are market value
- EBS liabilities are fair value: BEL + Risk Margin
- BEL uses discount rates based on either:
  - Standard approach: Based on market-representative portfolio
  - Scenario-based approach: Based on yields from company assets with adjustments
- Risk Margin: 6% CoC approach discounted at prescribed risk-free rates

Outline the BSCR risk categories.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

1. **Market risk**
   - Fixed income/equity risk: Prescribed factors $\times$ asset MV
   - Interest rate risk (1 of 2 approaches):
     1.1 Duration-based: ($|$Asset Duration $-$ Liability Duration$|$) $\times$ 2% $\times$ Asset MV
     1.2 Shock-based: Prescribed interest rate shocks
   - Currency risk: Prescribed shocks varying by currency
   - Concentration risk: Double RC for 10 largest holdings from single issuer
2. **Long-term risk** – Risk from long-term liability fluctuations
   - Mortality risk: (Face $-$ BEL) $\times$ NAR factor
   - Stop loss risk: Prescribed % $\times$ premium
   - Morbidity risk: Prescribed % $\times$ premium
   - Longevity risk: Prescribed % $\times$ BEL
   - VA risk: Standard approach or CTE 95 internal model
   - Other long-term ins risk: Prescribed factor $\times$ BEL
3. **Credit risk** – Primarily counterparty risk
4. **P&C risk**

Describe Bermuda capital ratios and regulatory action levels.

Source: Regulatory-Capital-Adequacy: Regulatory Capital Adequacy for Life Ins. Cos.

$$\text{ECR Ratio} = \text{BSCR Ratio} = \frac{\text{Available Capital}}{\text{ECR}}$$

$$\text{TCL Ratio} = \frac{\text{Available Capital}}{\text{TCL}}$$

Regulatory actions determined by ECR Ratio:

Level	**ECR Ratio**	**Action**
Early Intervention Process	[100%, 120%)	Company submits remediation plan, provides additional reporting
Minimum Required Level	$<$100%	Company must improve capital position, faces wind-up

Why does nonforfeiture law exist?

Source: ILA101-105: Life and Annuity Non-forfeiture Practices

- Regulators want equitable treatment for persisters and lapsers
- Regulators want p/h to receive fair share of equity on lapse

List key pricing concepts/risks for nonforfeiture.

Source: ILA101-105: Life and Annuity Non-forfeiture Practices

- **Policies must provide min nonforfeiture value by US law**
- **Lapse risks:**
  - High lapses on high CSV products
  - Low lapses on low/no CSV products (e.g., T-100, lapse-supported)
  - Disintermediation: High lapses when rates ↑, asset MVs ↓
- **Nonforfeiture option utilization rates**
- **Anti-selection**: Mortality ↑ for options maximizing benefits/cost
- **Expense assumptions**: Differ from regular policies, vary by option
- **Substandard risks:**
  - May have same CSV as standard
  - Impact of later qualifying for standard

Compare RPU vs. ETI nonforfeiture options.

Source: ILA101-105: Life and Annuity Non-forfeiture Practices

1. **Reduced Paid-Up (RPU)**
   - Lower face than original: $\text{CSV}_{[x]+t} = (\text{RPU Face})A_{[x]+t}$
   - Maximizes coverage period
   - Low anti-selection

2. **Extended Term Insurance (ETI)**
   - Maximizes p/h face
   - ETI expenses $<$ RPU
   - Policy terminates when CSV = 0
   - **Higher anti-selection** (unless ETI is default)

List nonforfeiture options besides RPU and ETI.

1. **Automatic Premium Loan (APL)**
   - Preserves original contract, benefits, riders
   - P/h can repay loan anytime
   - Policy terminates if Loan Balance $>$ CSV
   - May entice p/h to stop paying premium
   - DB = Face $-$ Loan Amount
2. **Annuitization**
   - Convert CSV $\rightarrow$ payment stream (mostly DAs)
3. **Cash Surrender**
   - Equivalent cash value of nonforfeiture options
4. **Flexible UL Premium Payments**

Describe prospective approach for calculating nonforfeiture values.

Source: ILA101-105: Life and Annuity Non-forfeiture Practices

- **Forward-looking PV** based on expected future experience
- Maintains parity between persisters/lapsers

$$\text{NF Value} = \text{PVFB} + \text{PVFE} + \text{PV(Profit Contribution)} - \text{PV(Gross Prems)}$$

- **Net premium approach** (common for traditional US products):

$$\text{NF Value} = \text{PVFB} - \text{PV(Level Net Premium)}$$

$$\text{Level Net Premium \%} = \frac{\text{PV}_0\text{(FB)} + \text{PV}_0\text{(AcqCosts)}}{\text{PV}_0(\text{Gross Premiums})}$$

- Assumptions for future experience:
  - Persister approach: Mortality only decrement
  - Whole contract approach: Reflects p/h option to terminate/convert

Describe retrospective nonforfeiture approach.

Source: ILA101-105: Life and Annuity Non-forfeiture Practices

- **Backward-looking** accumulation value:

$$\begin{aligned}\text{NF Value} &= \text{Fund}_t - \text{Surrender Charge}_t \\ \text{Fund}_t &= \text{Fund}_{t-1} + \text{Prem}_t + \text{Interest}_t - \text{Ben Charges}_t - \text{Exp}_t - \text{Profit}_t\end{aligned}$$

- No-lapse guarantee: Policy stays in force even if NF Value < 0

Compare prospective vs. retrospective approaches.

Source: ILA101-105: Life and Annuity Non-forfeiture Practices

- **Product types (by premium flexibility):**
  - Prospective: Works well for fixed premium products (WL, etc.)
  - Retrospective: Works well for flexible premium products (UL, annuities, etc.)
- **Retrospective easier to explain** to p/h
  - Based on actual experience, not future assumptions
- **Prospective maintains more parity** between persisters/lapsers
- **Prospective NF value changes if assumptions change** (hard to explain)
- Prospective NF value depends on policy class definitions

Compare US vs. non-US nonforfeiture practices.

Source: ILA101-105: Life and Annuity Non-forfeiture Practices

1. **United States**
    - Permanent policies must have nonforfeiture value
    - Balance different views on equity:
        - Lapsers: Want max value
        - Persisters: Want to pay as little as possible for lapsers
        - Company: Wants compensation for risk/costs
        - Agent: Wants compensation for sale (AcqCosts)
        - Regulator: Maximize p/h return without threatening solvency
2. **Outside United States**
    - Unlike US, most countries do NOT have specific nonforfeiture laws
    - Some countries (e.g., Canada) may file general NF plan with regulator
    - United Kingdom:
        - Most life products are variable
        - Strong disclosure regulation

How do NGEs differ from policyholder dividends?

Source: Overview of NGEs

Principal NGE characteristics:

1. Affect cost of contractual benefits/policy values
2. Not guaranteed except for min/max values
3. Insurer can change unilaterally (subject to contractual/legal restrictions)

NGE Examples: COI charges, credited rates, index parameters

NGEs $\neq$ p/h dividends:

- P/h dividends = par p/h share of par block profits
- NGEs based on future experience expectations
- P/h dividends based on actual past experience

List 4+ NGE examples.

Source: Overview of NGEs

All NGEs subject to max/min values under contract/regulation

- **Credited rates**: Adjustable based on investment performance
- **Premium loads and other expenses**
- **COI charges**: Vary by UW class, sex, age, policy year
  - Guaranteed max rates comply with SNL/tax law
  - Guaranteed rates may be higher if anticipated mortality $>$ SNL mortality
- **Index parameters**: Adjustable caps/participation rates
- **Nonguaranteed bonuses**: AV credits or charge refunds under certain conditions
- **VA rider charges** for enhanced GLB riders
- **Indeterminate premiums**: Traditional products with guaranteed/nonguaranteed premium scales

List actuarial considerations for original NGE determination.

Source: Overview of NGEs

Actuary must balance profitability with consumer needs

- **Expected mortality**:
  - Varies with UW, target market, risk classes, policy sizes
- **COI charges relative to expected mortality**
- **Policy load/expense charges**:
  - How to cover policy issue expenses
  - Portion of premium covering costs
  - Revenue source for flexible premium products
- **Basis for credited interest**:
  - Whether investment yields cover credited interest/other costs
  - New money vs. portfolio method
- **Assess assumptions via sensitivity testing**:
  - Impact to profitability/competitiveness (individually and aggregate)

List actuarial considerations when revising NGEs.

Source: Overview of NGEs

NGE revisions allowed if experience doesn't align with AEFs set in original pricing

- **Insurers have faced lawsuits:**
  - Changes not allowed by contract
  - Changes calculated inappropriately
  - Changes recouping past losses
- **Trend vs. variance:**
  - Future NGE changes result from variances from expected values
  - NGE assumptions should already anticipate experience trends
    - If actual trend aligns with expected → no changes needed
- Insurers can provide illustrations of possible future values to p/h

Define the following from ASOP 2:

- NGE Framework
- Determination Policy
- Policy Classes

Source: Overview of NGEs

NGE Framework = combination of:

- Company's **determination policy**
- Method to define **policy class**
- Any other relevant criteria/principles for determining NGEs

Determination policy = insurer's governing principles/objectives for NGEs

- Could be single document or collection of documents
- Includes profitability/capital objectives/requirements
- Specifies NGE review frequency with max time between reviews

Policy classes = policies grouped together when determining NGEs

- Different NGEs can have different class definitions
  - Must appropriately reflect differences within AEFs
- Classes not expected to change after issue
  - Can be redefined if supported by new information

List ASOP 2 considerations for NGE scale determination/revision.

Source: Overview of NGEs

1. **Review prior determinations (including original)**:
   - If not available, use reasonable approach to reconstruct
   - Consider whether NGE scales consistent with policy language
2. **Analyze emerging experience vs. anticipated from prior determination**:
   - Expectation: Do <u>not</u> revise unless AEFs change
3. **Consider whether to recommend NGE scale revision**:
   - Not purely mathematical (consider multiple criteria)
   - Other stakeholders: Senior mgmt, marketing, legal, admin
4. **<u>If</u> doing revision, determine revised NGE scales**:
   - Should be purely <u>prospective</u>: Focus on future profitability
   - Do NOT recoup past losses or distribute past gains

List 3 ASOPs assisting with NGE data considerations.

Source: Overview of NGEs

- **ASOP 23** data considerations:
  - Actuaries must review data (or disclose why they didn't)
    - NOT required to audit data
  - Disclose data limitations and implications
- **ASOP 2** data considerations:
  - Use professional judgment if recreating past determinations
  - Disclose any reliance on others
    - Must still review data and methods/models using data
- **ASOP 25**: Guidance on credibility considerations

Describe actuarial guidance for NGE modeling.

ASOP 56: Guidance on modeling (developing, modifying, reviewing, etc.)

Ideally start with original or past determination model

If prior model not available/usable:

- Start with model designed for another purpose, modify it
- Example: CFT models for asset adequacy testing
    - May be done at aggregate level with simplifications
    - May include PADs not appropriate for NGE determination

Outline Model 582 requirements for basic illustrations.

Source: Overview of NGEs

Basic illustration: Sent to all applicants for policies subject to Model 582

- Must include numeric summary of DBs, policy values, premium outlay
- Must be created under 3 bases:
  1. Policy guarantees (least favorable)
  2. Insurer's **illustrated scale** (most favorable allowed)
  3. Insurer's illustrated scale with reduced NGEs (in between)
     - Based on avg of guaranteed values and illustrated scale values
- **Illustrated NGEs** cannot be more favorable than illustrated scale
  - Do not project NGE improvements
  - Label all NGEs "nonguaranteed"
- **Illustrated scale** $\leqslant$ lesser of:
  - Disciplined current scale (DCS)
  - Currently payable scale (current/actual scale)
- **Lapse-supported illustrations** NOT allowed

List Model 582 requirements for disciplined current scale.

Source: Overview of NGEs

DCS = scale based on insurer's <u>recent</u> historical experience

- Illustration Actuary (IA) must certify Model 582 compliance
- Reflect only insurer actions already taken
- Can NOT include projected experience improvements
- Assumed expenses $\geqslant$ minimums defined in Model 582

Model 582 points to **ASOP 24** for guidance on IA's work

- Defines parameters of DCS (NGE scale) certified annually by IA

DCS must also pass self-support test (SST):

- <u>Passed</u> if accumulated policy CFs $\geqslant$ illustrated CSV
- Must pass at every illustrated point starting at 15th anniversary
  - Second-to-die policies: Test 20th anniversary and later
- Cash flows based on DCS assumptions

Describe Model 582 Lapse-Supported Test.

Source: Overview of NGEs

Lapse-supported illustrations <u>not</u> allowed

- IA must certify illustrations are NOT lapse-supported
- Certification based on lapse support test
- **Lapse support test** = SST with modified lapse assumption:
  - Years 1–5: Same lapse rates as SST
  - Years 6+: Assume 100% persistency (0% lapses)

List provisions of NY Regulation 210.

Source: Overview of NGEs

- Effective March 2018 for policies issued in NY (inforce and new business)
- More prescriptive (strict) than ASOP 2
- Key requirements:
  - Max time between NGE reviews $\leqslant$ 5 years
  - Formally documented, board-approved NGE determination policy
  - Various specific reinsurance considerations
  - Waiting period for filings and disclosures to p/h

Describe regulation of NGEs in Canada.

Source: Overview of NGEs

ICA Requirements for NGEs:

- Insurer/board must set policies for dividends/bonuses and criteria for adjusting policy values
- Maintain separate par accounts
- Actuary reports annually to board on fairness of par management and adjustable criteria

CIA Edu. Note: Fairness Opinions

- Define experience factor classes at issue
- Apply policies consistently, based on objective methods
- Align with policy/board provisions; avoid unreasonable pooling/cross-subsidies
- Document support for fairness opinions

Describe regulation of NGEs in Asia.

Source: Overview of NGEs

Singapore

- MAS Notice 320: Internal governance for charges/expenses, par disclosures, board/senior mgmt/AA roles
- SoP L03: Guidance on fund policy management

Hong Kong

- Guideline 16: Fair treatment, AA advises board on p/h expectations (no prescribed NGE approach)
- "Green light process" for unit-linked: Benchmark charges, ensure "treating customers fairly"

Japan: One ASOP requires AA to confirm adequate funding of par policies

Describe regulation of NGEs in United Kingdom.

Source: Overview of NGEs

- Most current NGE regulation created in early 2000s
  - Primary focus: “TCF principle” (treat customers fairly)
  - Requires communication/disclosures to p/h around NGE determination
- Principles and Practices of Financial Management (PPFM) document
  - Published by UK insurers with with-profits (par) business
  - Outlined in “Financial Conduct Authority’s Conduct of Business Sourcebook”
  - Describes how NGEs are managed
  - Detail varies by insurer but generally high-level
  - Can be amended anytime with written notice to p/h

List data/information challenges for NGEs.

Source: Overview of NGEs

- **Original product pricing often unavailable or insufficiently detailed**
  - ASOP 2: OK to reconstruct if needed
- **Experience data considerations/challenges:**
  - Sufficient granularity of experience studies
  - Experience credibility
  - Whether anticipated experience factors supported by emerging experience
  - Materiality of experience changes
  - Sensitivity of profit to assumption changes
  - Availability of p/h behavior experience (premium persistency, lapse, etc.)
- **No industry standard for evaluating AEF changes**
  - ASOP 2: Changing NGE scales should NOT ↑ expected profit vs. original expectations

Formulas for product cash flow and solvency earnings?

Source: ILA101-104: Profit Measurement And Analysis

$$
\begin{aligned}
\text{ProdCashFlow}(t) &= \text{Prem}(t) - \text{Ben}(t) - \text{Exp}(t) \\
\text{PreTaxSolvEarn}(t) &= \text{ProdCashFlow}(t) + \text{InvIncome}(t) - \text{SolvResIncr}(t) \\
\text{AfterTaxSolvEarn}(t) &= \text{PreTaxSolvEarn}(t) - \text{Tax}(t)
\end{aligned}
$$

Formulas for distributable and stockholder earnings?

Distributable Earnings

$$\text{DistEarn}(t) = \text{AfterTaxSolvEarn}(t) - \text{ReqCapIncr}(t) + \text{ATInvIncRC}(t)$$

Stockholder Earnings (U.S. GAAP Earnings)

$$\begin{aligned}
\text{PreTaxStockEarn}(t) &= \text{ProdCashFlow}(t) + \text{InvIncome}(t) - \text{BenResIncr}(t) \\
&\quad - \text{DACAmort}(t) + \text{InvIncRC}(t) \\
&= \text{PreTaxSolvEarn}(t) + \text{SolvResIncr}(t) - \text{BenResIncr}(t) \\
&\quad - \text{DACAmort}(t) + \text{InvIncRC}(t) \\
\text{AfterTaxStockEarn}(t) &= \text{PreTaxStockEarn}(t) - \text{Tax}(t) - \text{TaxInvIncRC}(t) - \text{DefTaxProv}(t)
\end{aligned}$$

Formula for annual ROE?

Source: ILA101-104: Profit Measurement And Analysis

$$\text{Annual ROE}_t = \frac{\text{AfterTaxStockEarn}_t}{\text{EquityBase}_t}$$

$$\text{EquityBase} = \text{StockAssets} - \text{StockLiab}$$

$$\text{Stock Assets} = \text{SolvRes} + \text{ReqCap} + \text{DAC}$$

$$\text{StockLiab} = \text{BenRes} + \text{DefTaxLiab}$$

Alternatively, equity base can be mid-year:

$$\text{MidYearEquityBase}_t = \frac{\text{EquityBase}_t + \text{EquityBase}_{t-1}}{2}$$

List considerations for selecting profit goals.

Source: ILA101-104: Profit Measurement And Analysis

1. **Select accounting basis**:
   - Usually solvency (Statutory) $\Rightarrow$ distributable earnings
   - May also use stockholder basis (GAAP)
2. **Determine how to reflect risk**:
   - Formula-based: Function of risk (mortality, expense, investment)
   - Judgment-based
   - Sensitivity tests (recommended)
3. **Interest/discount rate considerations**:
   - Cost of capital = weighted avg of company's capital sources
   - Opportunity cost = return on alternative ventures with similar risk
   - Current/expected future capital position
   - Discounting use:
     - After-tax usually more appropriate than pre-tax
     - Use CoC if determining whether product produces acceptable return

Pre-tax earnings formulas for solvency and stockholder?

Source: ILA101-103: Taxes

$$
\begin{aligned}
\text{ProdCashFlow}(t) &= \text{Prem}(t) - \text{Ben}(t) - \text{Exp}(t) \\
\text{PreTaxSolvEarn}(t) &= \text{ProdCashFlow}(t) + \text{InvIncome}(t) - \text{SolvResIncr}(t) \\
\text{PreTaxStockEarn}(t) &= \text{ProdCashFlow}(t) + \text{InvIncome}(t) - \text{BenResIncr}(t) \\
&\quad - \text{DACAmort}(t) + \text{InvIncRC}(t) \\
&= \text{PreTaxSolvEarn}(t) + \text{SolvResIncr}(t) - \text{BenResIncr}(t) \\
&\quad - \text{DACAmort}(t) + \text{InvIncRC}(t) \\
&= \text{ProdCashFlow}(t) + \text{InvIncome}(t) - \text{EarnResIncr}(t) + \text{InvIncRC}(t)
\end{aligned}
$$

Compare timing vs. permanent differences.

Source: ILA101-103: Taxes

1. **Timing differences**: Differences in earnings that will eventually reverse
   - **Most common (usually most significant) example:**

$$\text{TimingDiff}(t) = \text{SolvResIncr}(t) - \text{TaxResIncr}(t)$$

   - Tax authorities prefer higher tax revenue (higher assets, lower liabilities)
2. **Permanent differences**: Differences that will not reverse
   - Example: Non-deductible investment income

$$\text{PermDiff}(t) = -\text{InvIncome}(t) \times \text{NonTaxInvPct}(t)$$

Formulas for taxable earnings and tax on earnings?

Source: ILA101-103: Taxes

$$
\begin{aligned}
\text{TaxableEarn}(t) &= \text{PreTaxSolvEarn}(t) + \text{TimingDiff}(t) + \text{PermDiff}(t) \\
&= \text{PreTaxSolvEarn}(t) + \text{SolvResIncr}(t) - \text{TaxResIncr}(t) + \text{PermDiff}(t) \\
&= \text{ProdCashFlow}(t) + \text{InvIncome}(t) - \text{TaxResIncr}(t) + \text{PermDiff}(t) \\
\text{Tax}(t) &= \text{TaxRate} \times \text{TaxableEarn}(t)
\end{aligned}
$$

How can negative taxable earnings affect profitability?

Source: ILA101-103: Taxes

At company level, total tax cannot be negative

- New products generate negative earnings
- Smaller insurers may not be able to deduct first year losses
  - Results in higher capital, lower ROI

Describe US tax treatment for capital gains/losses.

Source: ILA101-103: Taxes

- Capital gains: Immediately taxable
- Capital losses: Can only offset capital gains
- Capital losses can be carried forward to offset future capital gains
  - Creates timing difference
- If carryforwards not used within 3 years → expire
  - Creates permanent difference

Describe tax on “investment income less expenses.”

Attempts to tax annual ↑ in p/h value <u>and</u> insurer earnings

$$\text{InvIncome}(t) - \text{Exp}(t) = \text{Ben}(t) + \text{SolvResIncr}(t) - \text{Prem}(t) + \text{PreTaxSolvEarn}(t)$$
$$\text{IETax}(t) = \text{IETaxRate} \times \left[\text{InvIncome}(t) - \text{Exp}(t)\right]$$

Some countries may use this or use as minimum tax

NOT used in US or Canada

Describe tax on capital.

Source: ILA101-103: Taxes

Certain types of capital may be subject to tax (e.g., par surplus)

$$\text{TaxOnCap}(t) = \text{TaxCapital}(t) \times \text{CapTaxRate}(t)$$

Could be assessed in addition to tax on earnings or used as minimum tax

Describe purpose and function of DAC tax.

Source: ILA101-103: Taxes

- Temporarily $\uparrow$ taxable income as premiums collected
- Reversed over next 10 years
- DACTaxAmount(t) = additions to DAC tax $-$ deductions from DAC tax

$$\text{TaxableEarn}(t) = \text{PreTaxSolvEarn}(t) + \text{TimingDiff}(t) + \text{PermDiff}(t) + \text{DACTaxAmount}(t)$$

For each premium collected (any policy year) for life insurance:

Year	Add to Taxable Earnings	Deduct From Taxable Earnings
1	7.7%	5% of 7.7%
2−10	0	10% of 7.7%
11	0	5% of 7.7%

Describe tax treatment under US GAAP (stockholder) accounting.

Source: ILA101-103: Taxes

GAAP reports taxes on accrued basis

$$\begin{aligned}
\text{Tax}(t) &= \text{TaxRate} \times \text{TaxableEarnings} \\
\text{AccruedTax}(t) &= \text{TaxRate} \times \text{PreTaxStockEarn}(t) \\
\text{DefTaxProv}(t) &= \text{AccruedTax}(t) - \text{Tax}(t) \\
\text{AfterTaxStockEarn}(t) &= \text{PreTaxStockEarn}(t) - \text{AccruedTax}(t) \\
&= (1 - \text{TaxRate}) \times \text{PreTaxStockEarn}(t) \\
\text{DefTaxLiab}(t) &= \text{DefTaxLiab}(t-1) + \text{DefTaxProv}(t) \\
&= \text{TaxRate} \times [\text{TaxRes}(t) - \text{EarnRes}(t)]
\end{aligned}$$

Causes taxes to be level % of pre-tax stockholder earnings

DefTaxLiab = liability (or asset if negative) for future tax payments

Describe 4 accounting-based profitability indicators:

1. Book Value Per Share
2. Price-to-Book Ratio
3. ROE
4. Operating Margin

Source: ILA101-102 – Understanding Profitability in Life Insurance

1. **Book Value Per Share** = $\dfrac{\text{Book Value}}{\text{Total Shares}}$

2. **Price-to-Book Ratio** = $\dfrac{\text{Share Price}}{\text{Book Value Per Share}}$

3. **ROE** = $\dfrac{\text{Shareholder Profits}}{\text{Shareholder Equity}}$
   - Advantages: Well understood, common
   - Disadvantages: Meaningful only at portfolio level

4. **Operating Margin** = $\dfrac{\text{Operating Profits}}{\text{Net Premiums}}$
   - Advantage: Easy comparisons
   - Disadvantages: Doesn't reflect timing, CoC, or riskiness

Describe 3 accounting-based profitability indicators:

1. Benefits Ratio
2. Lapse Ratio
3. General Expense Ratio

Source: ILA101-102 – Understanding Profitability in Life Insurance

1. **Benefits Ratio** $= \dfrac{\text{P/h Benefits} + \Delta \text{ Reserves}}{\text{Prems + Other Charges and Income}}$
   - Advantages: Measures mortality/morbidity; easy to understand
   - Disadvantages: Short-term focus is misleading
2. **Lapse Ratio** $= \dfrac{\text{Lapsed Policies or Face}}{\text{In-Force Policies or Face}}$
   - Advantage: Indicates net persistency
   - Disadvantages: May not accurately reflect timing
3. **General Expense Ratio** $= \dfrac{\text{General Operating Exp}}{\text{Net Prems}}$
   - Advantages: Cross-company comparisons
   - Disadvantages: Meaningful only for comparing similar business

Describe 2 accounting-based profitability indicators:

1. Return on Assets (ROA)
2. Net Investment Result

Source: ILA101-102 – Understanding Profitability in Life Insurance

1. **Return on Assets (ROA)** = $\dfrac{\text{Profit}}{\text{Total Company Assets}}$
   - Advantages: Easy to calculate; common for DAs
   - Disadvantages: Not good for life products; may be volatile
2. **Net Investment Result** = Inv Gains $-$ Impairments
   - Good for products where this is important earnings source

List disadvantages of accounting-based indicators.

Source: ILA101-102 – Understanding Profitability in Life Insurance

- Statutory focus is solvency, not income
- Rules vary throughout world
- Volatility of GAAP and IFRS earnings
- Poor at comparing new and existing business
- Not usually meaningful at product level
- Assumptions are locked in
- No accounting for CoC

Describe MCEV (market-consistent embedded value).

Source: ILA101-102 – Understanding Profitability in Life Insurance

MCEV = VIF + Required Capital + Free Surplus

$$\begin{aligned}\text{VIF} &= \text{PV Future Profits} - \text{Value of Options and Guarantees}\\ &\quad - \text{Cost of Non-Hedgeable Risks} - \text{Frictional Cost of Req Cap}\\ \text{Req Cap} &= \text{MV of Cap Allocated to Support Business}\\ \text{Free Surplus} &= \text{MV of Any Extra Allocated Cap}\end{aligned}$$

- Assets based on trading value
- Liabilities valued using replicating portfolios
- VIF highly sensitive to assumption changes
- More downside interest rate risk (asymmetric)
- Mortality and morbidity risk can be hedged (symmetric)
- May be trumped by emerging standards (Solvency II, IFRS)

List components of MCEV earnings analysis.

Source: ILA101-102 – Understanding Profitability in Life Insurance

1. New business
2. Unwinding of existing business
3. Experience variances and assumption changes
4. Economic variances

Describe 3 MCEV-based profitability indicators:

- MCEV
- MCEV Earnings Analysis
- Value of In-Force (VIF)

Source: ILA101-102 – Understanding Profitability in Life Insurance

1. **MCEV**
   - Advantages: Measures long-term value; harmonized
   - Disadvantages: May be volatile or not understood by investors
2. **MCEV Earnings Analysis**
   - Advantages: Understand MCEV drivers
   - Disadvantages: Can be volatile
3. **Value of In-Force (VIF)**
   - Advantages: Shows BV net of relevant costs
   - Disadvantages: Can be volatile

Describe 3 MCEV-based profitability indicators:

- Value of New Business (VNB)
- New Business Margin
- Return on MCEV

Source: ILA101-102 – Understanding Profitability in Life Insurance

1. **Value of New Business (VNB)**
   - Advantages: Understand value of NB
   - Disadvantages: May be overly optimistic
2. **New Business Margin** $= \dfrac{\text{VNB}}{\text{NB Prems}}$
   - Advantages: Compare to expected profit margins
   - Disadvantages: May be overly optimistic
3. **Return on MCEV** $= \dfrac{\text{MCEV Earnings}}{\text{MCEV}}$
   - Advantages: Complements ROE
   - Disadvantages: May be overly optimistic

List primary earnings sources and risk factors for insurers.

Source: ILA101-102 – Understanding Profitability in Life Insurance

1. **UW Results** – mortality, health, lapse products
   - Worse-than-expected claims or surrenders
   - Adverse selection and moral hazard
   - Frequent product churning
2. **Inv Results** – savings products
   - Insurer's asset allocation
   - Market performance
   - Level of guarantees
   - Inv expertise
   - P/h loss sharing
   - Duration mismatch
3. **Inv Mgmt Fee Income** – unit-linked products
   - Market volatility
   - P/h behavior

How are life insurance DBs treated for tax purposes?

Source: ILA101-108: LIME Ch. 1

Full DB is tax-free under §101(a) if there is an insurable interest

- Insurable interest may not exist for
  - Employer-owned life insurance (COLI)
  - Transfers for value (e.g., life settlements)
- If DB is annuitized, the DB proportion of payments (level %) is not taxed

$$\frac{\text{DB}}{\sum \text{Expected Annuity Payments}} = \text{Non-Taxable \% of Each Payment}$$

Define lifetime distribution.

Lifetime distributions result when CSV is brought outside the contract before death

- Surrenders (partial or full), policy loans, dividends, maturities

Describe tax treatment of inside buildup.

Source: ILA101-108: LIME Ch. 1

7702 and 7702A determine how the inside buildup is taxed before death

- Never taxed on a 7702-compliant contract if CSV stays inside through death
- If a distribution occurs, taxation depends on MEC status (7702A)
  - Non-MECs: FIFO (best) – cost basis comes out first
  - MECs: LIFO (worst) – gains come out first
    - Additional 10% penalty tax on the gain
    - Policy loans trigger gains just like surrenders (never for non-MECs)
    - Taxed portion of policy loan $\uparrow$ inv in the contract
- Gain inside the contract = 7702 CSV $-$ Inv in the Contract
  - Inv in the Contract = Prems Paid $-$ Previous Distributions Excl From Income
  - CSV ignores surrender charges and policy loans
  - Actual surrender charges incurred are netted out of policy gain

Describe tax treatment of:

- Premiums paid and interest paid on policy loans
- 1035 contract exchanges
- Contract sales
- Gifted life insurance
- Deduction of loss on surrender or sale

Source: ILA101-108: LIME Ch. 1

- **Prems paid and interest paid on policy loans** – not tax deductible
- **1035 contract exchanges** – does not trigger taxable gain
  - Cost basis carries over
  - Works for any combination of life, annuity, or QLTCI contract exchanges
- **Contract sales** (e.g., life settlements)
  - Seller's taxable gain = price received − cost basis
  - Cost basis = prems paid − cumulative COI
  - Buyer's cost basis = price paid + future prems − future untaxed distributions
- **Gifted life insurance** – always tax-free
- **Deduction of loss on surrender or sale** – not tax deductible

How did DEFRA and TAMRA affect life insurance taxation?

Source: ILA101-108: LIME Ch. 1

DEFRA (1984): §7702 – applies to all contracts entered after 1984

- Requires a contract to be life insurance under "applicable law"
- Set definitional limitations (actuarial tests) to limit inv orientation
  - Contracts must satisfy CVAT or GPT / CV Corridor Test
  - Limits CSV relative to DB and/or prems paid

TAMRA (1988): §7702A – applies to contracts entered after 6/20/1988

- Defined the MEC, which mainly prevented abuse of UL as a tax shelter
  - MEC = 7702 contract that fails 7702A's 7-pay test
  - Taxation of distributions is much less favorable than non-MEC
- Amended 7702's rules for mortality charges

List classes of life insurance under §7702 and §7702A.

Source: ILA101-108: LIME Ch. 1

1. **Non-MEC Life Insurance Contracts**
   - Passes 7702 <u>and</u> 7702A
   - DB is tax-free and tax on inside buildup is deferred
   - Inv comes out first when distributed (FIFO)
2. **MECs: Investment-Oriented Contracts**
   - Passes 7702 but fails 7702A
   - DB is tax-free and tax on inside buildup is deferred
   - Gains come out first when distributed (LIFO)
   - Additional 10% penalty unless age $\geqslant 59\frac{1}{2}$ or disabled
   - Policy loans taxed
3. **Failed Contracts: "Short-Term Endowments" (Rare)**
   - Fails 7702
   - NAR portion of DB is tax-free
   - Inside buildup taxed annually
   - Insurer's tax reserve deduction is limited

What is §7702's applicable law requirement?

Requires a contract to be life insurance by law in the jurisdiction where issued

- Includes state law or foreign law
- Canada is the only other country that defines life insurance for tax purposes
- FATCA informs IRS about foreign-issued contracts owned by US taxpayers

7702 requires insurable interest

- P/h has an interest in the continued life of the insured
- Must be present for tax-free DBs

Define tax law terms:

- Statutes
- Regulations
- IRS pronouncements
- Legislative history
- Case law
- Private letter rulings

Courts ruling 7702/7702A issues will act in the following order

1. Look at the relevant **statute** (DEFRA, TAMRA, etc.)
2. Look at **regulations** if any exist
   - **IRS pronouncements** are official IRS interpretations of the Code
   - Generally have same effect as law
   - Includes **notices**, **revenue rulings**, and **revenue procedures**
3. Look at the statute's **legislative history**
   - **Formal legislative history** – formal congressional reports and colloquies
   - **Blue books** – bigger collection for each law but not as binding
4. Follow any binding **case law** – legal precedents established by courts

Private letter rulings – IRS decisions on situations related to a specific taxpayer

- Indicates IRS's thinking but have no precedential value

Describe CVAT (Cash Value Accumulation Test).

Source: ILA101-108: LIME Ch. 2

Policy passes if: $\text{CSV}_t \leqslant \text{NSP}_t$ <u>at all durations</u>

- **Prospective test:** must pass by the **terms of the contract**
  - Contract must be designed to pass now and in the future
- $\text{NSP}_t = \text{PV}_t(\text{DB}) + \text{PV}_t(\text{Endow}) + \text{PV}_t(\text{QAB Charges})$
  - Interest Rate = max[4%, Guaranteed Rate in Contract]
  - Mortality assumption: use 100% prevailing CSO (Notice 2006-95 safe harbor)
- CSV = CV <u>ignoring</u> SCs, policy loans, & reasonable termination dividends
  - CSV also equals any amount p/h can borrow against
  - Pricing consideration: remove as much uncertainty around CSV as possible

Describe Guideline Premium / Cash Value Corridor Test.

Must satisfy BOTH GPT and CV Corridor Test
Guideline Premium Limitation: Prems Paid $\leqslant \max(\text{GSP}, \sum \text{GLPs})$

- **Retrospective test:** requires measurement of actual events
- **Prems paid** = prems actually paid into the contract less:
  - Non-taxable distributions (inv portion)
  - Force outs (prems returned within 60 days to ensure GPT compliance)

Cash Value Corridor Test: $\text{DB}_t \geqslant \text{CSV}_t \times \text{Corridor Factor}_t$

- Set by law: Ages 0–40 factor = 250%; for ages 41–95, factors grade to 100%

Describe guideline premium limitation and calculation.

Source: ILA101-108: LIME Ch. 2

Prems Paid $\leqslant$ **max(GSP, $\sum$GLPs)**;

$$\text{GSP} = \frac{\text{PV}(\text{DB}) + \text{PV}(\text{Endow}) + \text{PV}(\text{QAB Charges}) + \text{PV}(\text{Exp})}{1 - \text{PremLoad\%}}$$

$$\text{GLP} = \left(\frac{\text{PV}\big(\text{DB}\big) + \text{PV}\big(\text{Endow}\big) + \text{PV}\big(\text{QAB Charges}\big) + \text{PV}(\text{Exp})}{\ddot{a}_x}\right) \times \left(\frac{1}{1 - \text{PremLoad\%}}\right)$$

- Mortality assumption: use 100% prevailing CSO (Notice 2006-95 safe harbor)
- GSP interest rate = max[6%, Rate Guaranteed in Contract]
- GLP interest rate = max[4%, Rate Guaranteed in Contract]
- Expense assumption: use reasonable exp expected to be charged

List considerations for choosing a 7702 test.

Strengths of CVAT = Weaknesses of GPT / Corridor Testing

. . . and vice versa

Normally choose 1 method and use for contract's life

Reasons to choose CVAT

- Good for traditional products
- Easier to ensure prospective compliance
- Simpler (only 1 test)
- Requires much less record-keeping

Reasons to choose GPT and CV corridor test

- Better for flexible prem products (e.g., UL)
- Errors are less likely to affect all policies; easier to deal with

Describe 7-Pay Test.

Source: ILA101-108: LIME Ch. 2

Policy is <u>not</u> a MEC if. . .

$$\text{Amount Paid} \leqslant (\text{7-Pay Prem}) \times t$$

$$\text{7-Pay Prem} = \frac{\text{PV(Future Benefits)}}{\ddot{a}_{x:\overline{7}|}} = \frac{\text{CVAT NSP}_0}{\ddot{a}_{x:\overline{7}|}} \quad \text{in most cases}$$

- Interest, mortality, and future benefits: same as CVAT
  - Policies designed to be funded at 7-pay limit usually pass CVAT
- Exclude exp (same as CVAT)
- If DB $<$ 10,000, add \$75 to 7-pay prem
- Aggregation rule – combine all policies on same insured when testing

Describe 7702 treatment of:

- Multi-year interest rate guarantees
- Policyholder dividends
- Interaction with SNL
- Net rate plans

Source: ILA101-108: LIME Ch. 2

- Interest rate guarantees
  - Include any initial interest guarantees
  - Include any guaranteed bonus interest
  - Disregard short-term guarantees after first year
- Interest rates assumed do <u>not</u> include p/h dividends
  - P/h dividend = ANY nonguaranteed element (IRC §808)
- SNL and product design challenges
  - Nonforfeiture interest rates have fallen steadily since 1980s
  - The 4% and 6% 7702/7702A minimums have not changed
- Net-rate plans: guaranteed interest rate should include implied mortality charges

Describe history leading to current mortality assumption regulation for future benefits.

Current safe harbor = 100% 2001 CSO

Contracts Issued 1988 or Later (History of Changes):

- TAMRA's "reasonable" mortality standard created confusion
  - Permanent mortality rule: max mortality = 100% prevailing CSO table
  - Interim mortality rule said "reasonable" meant "expected to be imposed"
- Notice 88-128 safe harbor = 100% of sex-distinct 1980 CSO
- **Notice 2006-95 safe harbor = 100% 2001 CSO (M/F SM/NS)**
  - Same mortality as current SNL
  - In effect for all post-2008 contracts
  - Grandfathers 1980 CSO contracts unless there are unusual changes

Contracts Issued Before 1988: Use mortality guaranteed in contract

How are expense charges regulated by 7702?

“Expense charges” refers to QAB charges and guideline prem exp

For contracts issued after 1988:

- Include exp that company reasonably expects to <u>actually charge</u>
- Use actual exp even if $<$ guaranteed

List 4 computational rules under 7702.

Source: ILA101-108: LIME Ch. 2

1. **Assumed DBs cannot increase**
   - Option 1 NAR should $\Downarrow$ through maturity
2. **Deemed maturity must be age 95–100**
   - Even if actual maturity age < 95
   - Even if ending mortality table age > 100
   - Limits inv orientation (no short-term endowments)
3. **Assume DBs until deemed maturity**
   - Allows partial endowments before age 95
4. **Assumed endowment $\leqslant$ smallest future DB**
   - "Endowment" includes CSV at maturity
   - Prohibits endowments $>$ DB

What are alternate DB rules under 7702?

Alternative rules "override" the non-increasing DB rule

1. **Increasing Death Benefits** (e.g., option 2 UL contracts)
   - Increasing DBs are allowed in the GLP calculation only
   - The increases must only be used to keep the NAR from decreasing
   - Guideline limitation = greater of:
     - GSP assuming a non-increasing DB
     - $\sum$ GLPs assuming an increasing DB
   - **Least endowment rule** – restricts option 2 endowment to be $\leqslant$ specified amount
2. **Net Level Reserve Test**
   - Allows a net level reserve test in place of CVAT

$$\text{CSV}_t \leqslant \text{Net Level Reserve}_t$$

   - Based on a NL prem paid through age 95+ and assumes NAR does not increase
   - Not well suited for contracts that allow face adjustments

What QABs are allowed under 7702, and how are they reflected in definitional limitations?

QABs Specifically Listed Under 7702

1. Guaranteed insurability riders
2. Accidental death and disability (AD&D) riders
3. Family term coverage – “family” means base insured, spouse, or child
4. Disability waiver – waiver of prems or monthly deductions (UL)

Include PV(QAB Charges) in CVAT NSP, GSP, GLP, and 7-Pay

- “Charges” = actual charges expected
- Effectively raises the limitations to allow for higher prems paid due to the QAB

Describe adjustment rules under 7702.

Source: ILA101-108: LIME Ch. 2

Changes are generally allowed if

- Initiated by p/h (e.g., DB change)
- Allowed by operation of the contract (e.g., PUA purchase)
- NOT initiated by company

Contractual changes have 2 potential effects

1. **May require recalculation of definitional limits**
   - CVAT is "self-adjusting" – recalculate based on new current DB
     - Still must pass prospectively
   - Guideline prems are more restricted
     - Recalculate GSP and GLP as of current attained age
     - Accumulate new GLP going forward
2. **May result in loss of grandfathered status**
   - Example: a 1980 CSO contract could become 2001 CSO
   - Usually not an issue as long as change is allowed in contract

List 4+ adjustments allowed under guideline prem test.

Source: ILA101-108: LIME Ch. 2

1. Changes in DB or endowment benefits requested by p/h
2. A DB change by operation of the contract that was not reflected due to computational rules
   - Example: can’t assume increasing DBs, but p/h might do it anyway
3. Purchase of PUAs
4. Addition/deletion of QAB
5. Change in DB option ($1 \leftrightarrow 2$)
6. Change in UW class or rating

What rules govern adjustment events for 7-Pay Test?

1. **Reduction-in-benefits rule**
   - Must recalculate 7-pay prem from issue if DB reduced during first 7 years
   - **Retroactive** – assume the contract always had the lower DB
   - Same for scheduled and unscheduled changes
   - If amount paid $>$ new 7-pay limitation $\Rightarrow$ MEC
   - Best practice: if a DB reduction is scheduled, reflect from the outset
   - Last-to-die contracts: rule applies even beyond year 7
2. **Material change rule** (for events other than DB reductions)
   - Triggered after any change to benefits or terms used in original 7-pay
   - Applies any time during life of the contract (even beyond 7th year)
   - Start a new 7-year test period using a new, “adjusted” 7-pay prem:

$$\text{7-pay}^{(\text{Adj})} = \frac{\text{NSP} - \text{CSV}}{\ddot{a}_{x:7}}$$

How is option 2 GLP calculated?

Source: ILA101-108: LIME Ch. 2

Option 2 DB = Face + Cash Value

$$A_x^{\text{DBO2}} = \left(\sum_{t=1}^{100-x} v^t q_{x+t-1}\right) + v^{100-x}$$

$$\text{GLP}_x^{\text{DBO2}} = \frac{\text{Face} \cdot A^{\text{DBO2}} + \text{PV(Exp)}}{\ddot{a}_{\overline{100-x}|}} \times \left(\frac{1}{1 - \text{PremExp\%}}\right)$$

Exp and QAB charges should also be discounted with interest only

CVAT NSP, GSP, and 7-pay always use the option 1 approach

What approaches can be used to calculate monthly mortality and COI rates?

Source: ILA101-108: LIME Ch. 2

- **Monthly Mortality Rates** – Can be arithmetic or exponential

$$q^{(m)} = \begin{cases} q/12 & \text{if arithmetic} \\ 1 - p^{1/12} & \text{if exponential} \end{cases}$$

  - Practically equal until age 55, then exponential grows much higher
  - Arithmetic approach produces lower option 1 prems
  - Exponential produces unreasonably large prems for option 2 contracts

- **COI Rates** – Good for projection-based methods (UL)

$$\text{COI} = \begin{cases} \dfrac{q/12}{1 - q/12} & \text{if arithmetic} \\ \dfrac{1 - p^{1/12}}{p^{1/12}} & \text{if exponential} \end{cases}$$

It is important to use the correct rate (COI Rate $>$ mortality rate)

Write basic commutation functions for option 1 and option 2 versions of A_x and $\ddot{a}_x$.

Source: ILA101-108: LIME Ch. 2

These are equivalent to basic actuarial principles:

$$A_x = \frac{M_x}{D_x} \qquad \ddot{a}_x = \frac{N_x}{D_x}$$

$$A_x^{\text{DBO2}} = \frac{M_x^{\text{DBO2}}}{D_x^{\text{DBO2}}} + v^{100-x} \qquad \ddot{a}_x^{\text{DBO2}} = \frac{N_x^{\text{DBO2}}}{D_x^{\text{DBO2}}}$$

Describe projection-based approach used under 7702 and 7702A.

Source: ILA101-108: LIME Ch. 2

Equivalent to basic actuarial principles approach, but more practical for UL

Basic idea: Solve for net level prem (NLP) by iteratively projecting cash flows

- For GLP, NLP = prem that matures contract at age 100
- For 7-Pay, NLP = prem that pays contract up at beginning of year 7

$$\mathrm{CV}_{t+1} = \left(\mathrm{CV}_t + \mathbf{NLP} - \mathrm{Exp}_t - \mathrm{NAR}_t \times \mathrm{COI\ Rate}_t\right) \times \left(1 + i_t^{(m)}\right)$$

What is immediate payment of claims adjustment?

Source: ILA101-108: LIME Ch. 2

To reflect that claims will be paid immediately:

$$A_x^{\text{IPC}} = \frac{i}{\delta} \cdot A_x$$

where $\delta = \ln(1+i)$ and $\frac{i}{\delta} \approx 1 + \frac{i}{2}$

Describe primary classifications of life insurance for tax purposes in Canada.

Source: ILA101-116: CIT Ch. 10: Taxation of Life Insurance Policies

Post-Dec 1, 1982 Policies

1. **Exempt policies ⇒ insurance-oriented (most favorable)**
   - Inside CSV build-up is only taxed on disposition
   - DBs are 100% tax-free
   - Must pass exempt test (different rules for pre-2017 vs. post-2016)
   - Burden is on insurer to maintain exempt status
2. **Non-exempt policies ⇒ savings-oriented (least favorable)**
   - CSV build-up is taxed annually ("accrual taxation")
   - Untaxed CSV build-up is taxed at death
   - Includes deferred annuities

Grandfathered Pre-Dec 2, 1982 Policies (**most favorable tax status of all**)

- No exempt/non-exempt distinction and higher cost basis
- If lose grandfathered status ⇒ deemed disposition ⇒ perform exempt testing

Describe Canadian exempt policy rules.

Source: ILA101-116: CIT Ch. 10: Taxation of Life Insurance Policies

"Existing" rules for pre-2017 issues, and "new" rules for post-2016 issue

- **If policy fails exempt test, it is non-exempt for life**
- **A coverage remains exempt if this is true for all anniversaries (t's):**

$$\mathrm{AF}_t \leqslant \mathrm{AF}_t^{\mathrm{ETP}}$$

- **AF = MTAR = maximum tax actuarial reserve**
  - Post-2016: MTAR = max(CSV Gross of SC, NPR)
  - Pre-2017: MTAR = max(CSV Net of SC, 1.5 PTR)
- **ETP = Exempt test policy**
  - Post-2016 ETP: 8-pay endowment at 90
    - Mortality = generally same as actual policy
    - Interest rate = 3.5%
  - Pre-2017 ETP: 20-pay endowment at 85
    - Mortality = generally same as actual policy
    - Interest rate = max(contract rate, 4%)

How is exempt test policy accumulating fund calculated?

Source: ILA101-116: CIT Ch. 10: Taxation of Life Insurance Policies

The AF grades linearly through the pay period, then equals PVFB

$$\text{Post-2016 AF}_t^{\text{ETP}} = \begin{cases} \text{DB}_t \times A_{[x]+8:\overline{90-[x]-8}|} \times \left(\dfrac{t}{8}\right) & \text{if } t \leqslant 8 \text{ and } [x]+t < 90 \\ \text{DB}_t \times A_{[x]+t:\overline{90-[x]-t}|} & \text{if } t > 8 \text{ and } [x]+t < 90 \\ \text{DB}_t & \text{if } [x]+t \geqslant 90 \end{cases}$$

$$\text{Pre-2017 AF}_t^{\text{ETP}} = \begin{cases} \text{DB}_t \times A_{[x]+20:\overline{85-[x]-20}|} \times \left(\dfrac{t}{20}\right) & \text{if } t \leqslant 20 \text{ and } [x]+t < 85 \\ \text{DB}_t \times A_{[x]+t:\overline{85-[x]-t}|} & \text{if } t > 20 \text{ and } [x]+t < 85 \\ \text{DB}_t & \text{if } [x]+t \geqslant 85 \end{cases}$$

List anti-avoidance rules under Canadian insurance taxation.

Source: ILA101-116: CIT Ch. 10: Taxation of Life Insurance Policies

These rules put additional limits on cash accumulation to prevent abuse

1. **Always use total DB for ETP face amount**
2. **8% rule:** If DB $\uparrow$ by $>$ 8% between anniversaries:
   - "Issue" new ETP for excess increase
   - Test excess DB's CSV against new ETP AF
   - Post-2016: applies at coverage level
   - Pre-2017: applies to total policy DB
3. **250% rule:** Fund growth capped at 250% every 3 years starting in year 10

$$\frac{\text{Policy AF}_t}{\text{Policy AF}_{t-3}} \leqslant 250\% \quad \text{for } t \geqslant 10$$

   Only applies if policy AF $> \frac{3}{20}$ of total ETP AFs
   - If triggered, set all ETP issue dates = max(actual issue date, $t-3$)
   - Will not apply again for 6 years once triggered

How is adjusted cost basis determined for a typical exempt life insurance policy in Canada?

Source: ILA101-116: CIT Ch. 10: Taxation of Life Insurance Policies

Generally speaking, for an **exempt** policy…

$$\text{ACB} = \sum \text{Prems} + \sum \text{Taxable POD} \ - \sum \text{NCPI} - \sum \text{Total POD}$$

- Prems = all prems paid except portion funded by a disposition
  - Also include amounts paid to acquire (e.g., life settlement)
- Taxable POD = portion of all dispositions included in income
  - ACB falls by Total POD $-$ Taxable POD
- NCPI = net cost of pure insurance

$$\begin{aligned} \text{NCPI}_t &= q_{x+t} \times (\text{DB}_t - \text{AF}_t^*) \\ q_{x+t} &= \text{prescribed mortality rate} \\ \text{AF}_t^* &= \text{Policy AF ignoring policy loans} = \text{gross CSV (usually)} \end{aligned}$$

List 4+ types of dispositions for Canadian tax purposes.

All of the following are considered dispositions:

1. Full and partial surrenders
2. Policy loans
3. Policy dividends
4. Maturity benefits
5. Absolute assignment of ownership (sale, transfer, or gift to another party)
6. If a previously exempt policy fails the exempt test
7. If a non-exempt policy dies

How is taxable income and ACB impact determined by pre-death dispositions in Canada?

Source: ILA101-116: CIT Ch. 10: Taxation of Life Insurance Policies

$$\text{Taxable Income} = \begin{cases} \text{POD} - \text{Prorated ACB} & \text{for post-Dec 1982 partial surrenders} \\ \max(\text{POD} - \text{Full ACB}, 0) & \text{for all other dispositions} \end{cases}$$

Post-Dec 1982 partial surrenders ONLY:

$$\text{Prorated ACB} = \begin{cases} \frac{\text{Partial Surrender}}{\text{CSV}-\text{Policy Loan}} \times \text{Current ACB} & \text{if post-2016} \\ \frac{\text{Partial Surrender}}{\text{Policy AF}} \times \text{Current ACB} & \text{if pre-2017} \end{cases}$$

Remember: pre-2017 policy AF = max(CSV Net of SC, 1.5 PTR), so it's not always the CSV!

In ALL Situations:

$$\text{ACB After} = \text{ACB Before} - \text{POD} + \text{Taxable Income}$$

How are policy loans taxed in Canada?

Policy loan POD = Loan Amount − Portion Used to Pay Prem

- Cannot exceed CSV − Prior Policy Loan Balance

Taxable income and ACB impact = same as any other non-partial surrender

Future ACB impacts:

- ADD non-deductible paid policy loan interest to ACB
  - Don't add capitalized interest
- ADD loan repayments to ACB after gain is repaid
  - Repaid taxable income from policy loans is deductible (so don't add it)
  - Net effect of repayments is to restore the original ACB reduction

How does accrual taxation work for non-exempt life insurance policies in Canada?

Post-Dec 1982 non-exempt policies are subject to accrual taxation

Post-1989 policies: every anniversary (and on death):

$$\begin{aligned} \text{Taxable Income}_t &= \text{Policy AF}_t - \text{ACB Before}_t \\ \text{ACB After}_t &= \text{ACB Before}_t - \text{POD} + \text{Taxable Income} \\ &= \text{Policy AF}_t \end{aligned}$$

List ways a policy can lose grandfathered status.

If any of these occur, grandfathered status is lost forever

- Prescribed increases in BOTH prems and DBs
  - “Prescribed” means “unscheduled” in contract
  - Exceptions for changes in u/w class, payment frequency, certain riders
- Prescribed prem paid on a non-exempt policy
- Non-permitted changes in ownership

The following will never result in loss of grandfathered status:

- Adding a term or other rider
- Non-arm’s length transfer (e.g., assign to a family member)
- Roll overs into an annuity contract

Describe financial advisor distribution channel.

Typical market: wealthy clients, more sophisticated products

- Individuals who arrived from outside of the life insurance industry
- May be found in inv advisory companies, banks, or credit unions
- Stockbrokers (largest group)
- Does NOT include fee-based planners
- Reached through wholesalers
- Most advanced uses of life insurance
- More face-to-face interaction than other channels

Describe independent agent distribution channel.

Typical market: middle to high income clients

- **Includes brokers and PPGAs**
- **Producer groups common**
  - Benefits for agents: higher comp, networking, education
- **Wholesalers** – how companies reach independent agents
  - Spread-sheeting model vs. assisted sales model
  - Can be organized by product line or one for all product lines
  - External vs. internal wholesalers
- Companies compete over agents with commissions and services provided

Describe the career agent distribution channel.

Typical market: middle to high income clients

- **Recruited and hired by a single company**, but often sell for other companies too
- Field sales offices vs. agencies
  - Agency Managers: salaried employees; company pays expenses
  - General Agencies: independent contractors; paid commissions/overrides; GA pays expenses; harder to replace
- Other staff: specialists and sales managers
- **Recent trends: Numbers have been declining**
  - Very expensive form of distribution
  - Agent retention has been low
  - Independent agents becoming more popular
  - Moving from smaller to “mega-agencies”
  - Traditional distinctions between agency managers and GAs blurring

Describe MLEA distribution channel.

Typical market: mostly middle income <u>life</u> clients

- **Captive** – Only sell only for one company
- Known for extremely high retention rates (80%, which is 4x career agents)
- Sell both P&C and life insurance
- P&C sales are very different from life insurance
    - P&C ownership is mandatory
    - P&C sales process is "one and done"
    - Wealthy MLEA clients more likely to be P&C
    - Life insurance requires more agent involvement during u/w process
- Life insurance clients: few wealthy clients; mostly middle income

Describe IMO distribution channel.

Typical market: low to middle income

- Independent agency-like organizations
- IMOs recruit, train, and support agents
- Differences with career agent system:
  - IMO system is built for high productivity
  - No business brokered outside of IMO
- Highly structured sales presentations
- Usually focus on specific set of products sold to a specific market

Describe call center distribution channel.

Typical market: low to middle income

- Prospects directed to toll-free number
- Effective with affinity groups and association marketing (e.g., college alumni)
- Gaining popularity

Describe home service agent distribution channel.

Typical markets: lower income, inner cities, rural south

- Personal, face-to-face interaction between prospect and sales intermediary
- Captive – only sell for one company
- Traditional form: industrial or debit life insurance
  - Agents collected prem door to door weekly
  - Commissions paid on growth, not individual sales
  - Became obsolete (labor force changes, selling practices, agent safety)
- Monthly debit ordinary has replaced weekly industrial prem collection

Describe worksite distribution channel.

Typical market: low to middle income, smaller face policies

- Employers allow policy prems to be paid via payroll deduction
- Intended to supplement employees' existing coverage
- Consumers like because
  - Ease of purchase and convenience
  - Perception that employer vetted quality first
- Insurer must be an expert in administrative and payroll systems

Describe forms of direct distribution channels.

Source: A&S Ch. 3: Primary Life Insurance Channels

Typical market: low to middle income

1. **Internet direct**
   - Company-specific or market-specific websites
   - Aggregators (most popular) – multiple companies on one site
     - Mostly term life with larger face amounts than direct mail
     - Companies compete on price
     - Simplified issue u/w
2. **Pure direct**
   - Mailed solicitations
   - More effective when combined with human intervention (e.g., sales rep in call center)
   - Overly saturated (not very effective)
   - Only 2–5% realized prem

List 4 main steps performed by a distribution channel.

Source: A&S Ch. 4: Functions of Distribution

1. Step 1: Prospecting for New Clients
2. Step 2: Needs Analysis
3. Step 3: Presenting and Closing
4. Step 4: Managing the Client Relationship

Describe how distribution channels prospect for new clients.

Source: A&S Ch. 4: Functions of Distribution

- **Traditionally the most important function**
- **Main reason agents fail:** don't generate enough prospects
- **Prospecting methods**
  - Cold calling
  - Direct mail
  - Event based networking
  - Social media
  - Referred leads
- **Ultimate goal:** word of mouth advertising
- Channel considerations
  - Call center and/or direct methods don't use prospecting
  - Traditional agents – Difficult to measure success
  - IMOs have clearest process (IMO gives leads to agents or agents pay)

Describe how distribution channels perform needs analysis.

Key steps:

1. Judge the likelihood that the prospect will buy
2. Begin a sales discussion

Variations

- Level varies with socio-economic status of client
- Varies with channel
  - Financial planners – more likely to do sophisticated analysis
  - IMOs – do little comprehensive analysis (specialized already)
  - Direct – may not do any at all (little or no face-to-face contact)

Describe key steps that distribution channels follow to present and close contracts.

Source: A&S Ch. 4: Functions of Distribution

1. **Producer presents illustrations**
    - Presented for different products, different companies
    - Regulated
2. **Agent initiates application and u/w process**
3. **U/w steps**
    - Long list of medical and lifestyle questions
    - Paramedical exam
    - Ways of speeding up the process: simplified issue, tele-u/w
    - Agents are MUCH less involved today
4. **Policy delivered in person by agent**

Describe role of distribution channels in managing client relationship and key challenges.

Source: A&S Ch. 4: Functions of Distribution

- P/h services is mostly done by home office
  - Frees agents' time to focus on selling and prospecting
- **Retention programs have mostly failed**
  - Agent retention is a major challenge
  - Orphans must be serviced by home office
  - Levelized commissions hurt new agents
  - Poor comp for distribution systems for POS
  - More focus on sales than retention
- **Agents still deliver DBs**

Describe forms of producer compensation:

1. Commissions
2. Financing
3. Commission advances
4. Vesting Schedules
5. Bonuses
6. Expense Reimbursement Allowance (ERA)
7. Benefits
8. Recognition

Source: A&S Ch. 6: Distribution Compensation

1. **Commissions**
   - Heaped, annualized commissions common
   - Claw backs mitigate FY lapses
   - ↑ renewal comm, ↑ persistency
2. **Financing**
   - Validation requirements
   - Financing plans – financial assistance for new agents
3. **Commission advances**
   - Debt owed to company
4. **Vesting Schedules**
   - Usually fully vests after a given number of years
5. **Bonuses (e.g., persistency bonus)**
   - Often % of prem
   - Not guar. but hard to take away
6. **Expense Reimbursement Allowance (ERA)**
   - Covers agent's legit business exp
   - Regulated
   - Behaves like a bonus
7. **Benefits (Career Agents)**
   - 401k, health insurance, etc.
8. **Recognition**
   - Incentive trips
   - Regulated
   - May be very expensive

Describe 2 forms of agency head compensation.

Source: A&S Ch. 6: Distribution Compensation

1. **Override** = % of agent commission
   - Traditional form of agency manager comp
2. **Gross Dealer Concession (GDC)**
   - Common for brokerage inv products
   - Total revenue paid to producer for selling
   - More transparent
     - Written in contract
     - Fully discloses to manager and salesperson
   - Salesperson gets a % of GDC
     - GDC grid – salesperson % $\uparrow$ with production
   - Approaches for life insurance
     - GDC = agent commission + overrides
     - GDC priced into exp margin

How does regulation limit distribution compensation?

Source: A&S Ch. 6: Distribution Compensation

- **NY State Regulation 4228**
  - Limits total spending on sales comp
  - Extraterritorial
- **SEC regulation**
  - Pertains to variable products

Describe 2 important pricing considerations related to distribution compensation.

Source: A&S Ch. 7: The Economics of Life Insurance Distribution

1. **Very large and front-loaded**
   - High first-year commissions
   - Large costs paid before sales are made
2. **Prem must cover distribution costs**
   - Don’t omit or double count
   - Include overhead: yes or no?

Describe 4 types of contingent or non-contingent distribution expenses.

Source: A&S Ch. 7: The Economics of Life Insurance Distribution

- **Not Contingent (given)**
  - Building rent and other fixed costs
- **Sales contingent**
  - Commissions and overrides
  - Recognition
  - Benefits
  - Production bonuses
- **Persistency-contingent**
  - Renewal commissions and overrides
  - Chargebacks
- **Agent longevity**
  - Vesting
  - Field mgmt comp

Describe 2 distribution expense modeling approaches.

Source: A&S Ch. 7: The Economics of Life Insurance Distribution

1. **Model contingencies within the pricing model**
   - More complex but easier to validate
2. **Enter as a single input**
   - Model contingent distribution costs separately first
   - Simpler but harder to validate

Describe how distribution loads can be priced into life insurance premiums.

Distribution loading approach

- 2 ways to start
  - Create standalone model of distribution channel
  - Distribution business unit provides initial estimate
- Assumes the market sets the prem
- Distribution load =
  - Prem remaining after benefits, exp, and desired profit
  - Distribution channel's revenue
- Distribution channel manages its exp within the load revenue
  - Clearer economic value
  - Channel can justify costs if lead to higher revenue
- No need to model contingent distribution costs

Which distribution channels are predicted to decline?

Source: A&S Ch. 10: The Future of Life Insurance Distribution

- **Direct Mail**
  - Decline due to consumer saturation and $\uparrow$ reliance on technology
  - Applies to <u>pure</u> direct mail without human interaction
- **Home Service**
  - Long-term decline with no signs of reversal
- **Career Agents** and **Independent Agents**
  - Decline due to aging workforce and difficulty attracting younger agents

Which distribution channels are predicted to be stable?

Source: A&S Ch. 10: The Future of Life Insurance Distribution

- **Internet Direct**
  - Consumers prefer personalized advice and human interaction
  - Technology will influence shopping behavior
  - Uncertainty: Will Gen Y desire human interaction like prior generations?
- **Voluntary Worksite Marketing**
  - Growth has plateaued due to market saturation
  - Focus has shifted to selling specialty products (e.g., identity theft protection)

Which distribution channels are predicted to grow?

Source: A&S Ch. 10: The Future of Life Insurance Distribution

Predicted to have slow growth:

- **IMOs** and **MLEAs**
  - Slow, steady growth by targeting under-insured middle market
- **Financial Advisor**
  - Growth as agents offer broader financial services
  - Limited life insurance growth due to estate planning law changes

Predicted to have rapid growth:

- **Call Center-Mediated Direct Response**
  - Avoids negative reputation of traditional agents (e.g., overselling, pressure)
  - Combines personal interaction with efficiency, meeting consumer preferences

Describe recent trends in life insurance sales.

Source: Life Insurance for the Digital Age: An End-to-End View

- **Sales have been declining and are at a 50-year low**
  - Sharpest decline: middle-income and millennial market
  - Average US household is under-insured by 400K
- **Average face amount has risen**
  - Agents now focus on the higher net worth market
- **Distribution channels are not designed to serve the mid-market**

Describe traditional forms of u/w and how end-to-end u/w is different.

Source: Life Insurance for the Digital Age: An End-to-End View

Traditional u/w methods:

- **Full u/w (lowest mortality cost)**
  - Medical/para-medical exam
  - Collection of fluids (blood/urine)
  - Rx, MVR, and MIB
- **Non-medical u/w (higher mortality cost)**
  - No medical/para-medical exam
  - No fluid collection
  - Rx, MVR, and MIB

End-to-end approach:

- Supplements non-medical u/w with predictive analytics
- Can bring non-medical mortality cost down to fully u/w levels
- More customer friendly (less invasive)
- Faster, cheaper

List 4+ ways that insurers can improve customer satisfaction.

Source: Life Insurance for the Digital Age: An End-to-End View

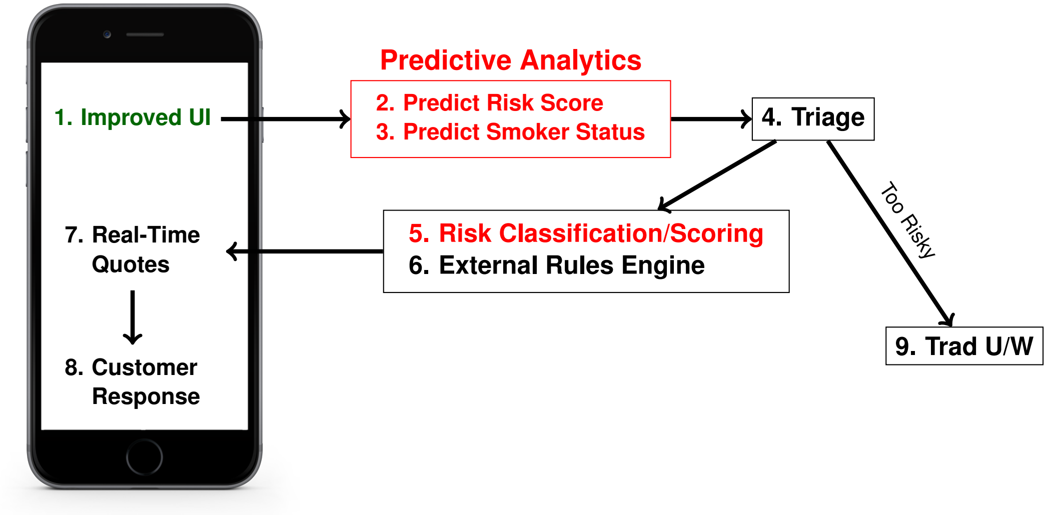
Insurance customer satisfaction $<<$ personal banking and retail

How to $\uparrow$ mid-market and millennial customer satisfaction:

1. **Quick tutorial** on life insurance with concise explanation of benefits
2. **Individualize needs analysis** and recommendations
3. **Simpler application process** (fewer questions and pre-filled info)
4. **Real time quotes** similar to auto insurance
5. **Relevant quote alternatives** for comparison
6. **View of life insurance and related products** (e.g., purchased by peers)

Review end-to-end u/w process on back of card. Be able to describe each step.

Source: Life Insurance for the Digital Age: An End-to-End View



List key trends with life insurance resulting from low interest rates

Source: Market Trends…in a Rising Interest Rate Environment

- Prem rates ↑, especially for term
- Term and WL sales remained steady
- Multiple companies plan to exit the life insurance completely (or already have)
- Major ULSG carriers have paused ULSG sales
- Active mgmt of NGEs (e.g., raised COI charges) even if it means litigation
- Insurers want to ↑ retention of profitable post-level term business

List key trends with annuities resulting from low interest rates

Source: Market Trends…in a Rising Interest Rate Environment

- More focus on accumulation; less on long-tailed secondary guarantees
- RILA market has expanded
- VA market share has ↓; some insurers have exited
- More GA product competition from PE-backed companies
- ↑ use of reinsurance for fixed annuities, including FIAs
  - Redundant reserve financing treaties utilized to aid profitability
  - Offshore reinsurance used for ALM and access to alternative inv
- Fixed annuity sales ↑ when credit spreads ↑
- Active mgmt of NGEs (crediting rates)

List key trends with LTC insurance resulting from low interest rates

Source: Market Trends…in a Rising Interest Rate Environment

- Carriers continue to exit the standalone LTC market
- Features of new standalone LTC products:
  - Simpler designs
  - Less rich benefits
  - Cost-sharing features (e.g., co-pays)
- Some mutual companies have introduced par LTC policies
- $\uparrow$ in LTC combo products (e.g., base life policy + LTC rider)
  - New product chassis mitigate interest rate risk (VUL)
- Continued inv in inforce mgmt, including prem rate $\uparrow$

Describe general trends in past 20 years when interest rates have risen and how current period may affect insurers

Source: Market Trends... in a Rising Interest Rate Environment

Past periods of rising interest rates have 2 things in common:

1. The 10-year rate only ↑ 1% to 1.5% over a 2-year period
2. Rates fell sharply after each period

Assuming the trends are repeated with the current rise in interest rates:

1. Portfolio rates won't change significantly, so it won't improve insurers' profits
2. The ↑ in rates may be short-term

Provide examples of product designs that perform best when interest rates rise

Source: Market Trends… in a Rising Interest Rate Environment

Focus on flexible product designs

- WL is the most stable when interest rates change dramatically
  - Simple, compatible product design
  - Use of portfolio pricing instead of new money
- Interest-sensitive products are hurt when interest rates are low
  - ULSG becomes expensive for p/h
  - FAs/MYGAs become unattractive compared to FIA, IUL, VAs
  - FIA/IUL perform well when rates rise if equity markets are also rising

Provide 4+ examples of products that perform well and mitigate interest rate risk along with advantages to insurers and p/h

Source: Market Trends…in a Rising Interest Rate Environment

- **1-year rate guarantee instead of multi-year**
  - Pro to insurer: can make timely adjustments based on market
  - Pro to p/h: potentially higher guarantees that adapt to markets
- **Multi-year rate guarantees with small stair-step rate $\uparrow$**
  - Pro to insurer: less risk since steps are small (e.g., 15 bps)
  - Pro to p/h: mitigates risk of low/flat guarantee
- **Participation in equity markets/indexes**
  - Pro to insurer: higher illustrated returns with lower guarantees
  - Pro to p/h: earn returns comparable to other financial products
- **Monthly cap instead of annual cap**
  - Pro to insurer: $\uparrow$ product popularity with low risk of large loss
  - Pro to p/h: possibility of much higher annual return
- **Cumulative guarantees over a defined period (e.g., 2%/year for 5 years)**
  - Pro to insurer: cheaper than annual guarantee
  - Pro to p/h: same guaranteed $\uparrow$ over a longer term
- **Participation by paying dividends during favorable market conditions**
  - Pro to insurer: weaker guarantee, so less risk in unfavorable markets
  - Pro to p/h: can participate in insurer's performance
- **Stacking rollup for FIA GLWBs**
  - Pro to insurer: $\downarrow$ strain and adds flexibility
  - Pro to p/h: potential upside if market performs well

Describe how a temporary rise in interest rates might affect insurer’s portfolio returns

If the rise in rates is short-term, portfolio rates will continue to fall

- Be prudent in providing higher credited rates and/or longer guarantees
- Reduce nonguaranteed benefits and costs (e.g., sales rep comp)
- Closely monitor features (e.g., ROP riders) where low interest rates with level prems will erode profits

Provide an example of a tail risk scenario involving rising interest rates

Source: Market Trends…in a Rising Interest Rate Environment

- Early 1980s: UL writers shortened asset portfolios when ST yields $>$ LT yields
- When ST yields fell, UL writers moved to risky assets to maintain returns
  - Worked well until junk bond market collapsed in the early 1990s
  - 3 large insurers went insolvent with mass UL surrenders